Informative Quantum Actions and Predictive Goal Geometry

Operational Meaning, Low-Dimensional Constructions, and an Exact Bellman Obstruction

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Abstract

Can an agent learn that its available quantum interventions mean "move north," "move east," and so forth, without those meanings being supplied as labels? This paper develops an operational framework in which an action is defined by the future outcome statistics it induces. Controlled quantum instruments generate histories; equivalence of histories under all future tests defines predictive states; and equivalence of actions is quotienting by every experimentally invisible realization gauge. Goal-directed control then equips predictive states with Bellman costs. A two-dimensional hodological space exists only when these costs pass metric and Euclidean embeddability tests, most sharply Schoenberg's centered-Gram criterion.

Two low-dimensional studies make the distinction between *information* and *geometry* concrete. A qubit instrument has an exactly invertible predictive phase chart and its two opaque actions are learned from present and future data, but its nonunitary dynamics fail path-independent composition. A qutrit Hesse-SIC instrument is more strongly integrated: every action both translates a hidden $\mathbb{Z}_3^2$ label and reports a noisy outcome. From shuffled tokens alone, maximum-likelihood transition tables recover the toroidal action topology exactly. Nevertheless, its undiscounted Bellman goal cost is 4 both at a goal and at an adjacent state, violating identity of indiscernibles. Weakening the measurement does not repair this equality. Separating motion from reporting gives the exact benchmark $V_g(s) = 6 + d_T(s, g)$, but no longer solves the integrated-action problem.

The resulting lesson is constructive but deliberately limited. Small Hilbert spaces can support learned operational action semantics and exact finite topology. They do not automatically produce a metric merely because their latent action graph is spatial. We prove a no-information-without-disturbance theorem under explicit universal assumptions, state finite necessary-and-sufficient criteria for predictive two-dimensional geometry, and propose a program based on weak covariant instruments, predictive atlases, and control-aware measurement design.

Contents

1 Why action meaning should be operational

The motivating hypothesis is that space need not be a primitive arena. For a goal-directed agent, a goal reached cheaply is near and one requiring many risky or costly interventions is far. A geometry reconstructed from such costs is *hodological*: it is a geometry of routes, obstacles, and attainability. If a family of goals admits a faithful two- or three-dimensional embedding, an agent's policy may be interpreted as motion through an emergent space.

There is an important circularity to avoid. If the experimenter names two buttons X and Y, declares that they translate horizontal and vertical coordinates, and scores the agent by distance on that declared grid, the spatial interpretation was inserted before learning. The stronger question is:

> Can the agent infer the operational meanings of opaque interventions from outcome statistics, then discover that its goal costs possess low-dimensional spatial structure?

The phrase *operational meaning* is essential. Quantum channels have many equivalent Kraus and coordinate descriptions. Only probabilities of possible future experiments can be learned. We therefore build the theory from histories and tests, not from a preferred matrix representation.

This paper separates four questions that are often conflated:

1. **Immediate informativeness:** does the outcome emitted by the current action depend on the current predictive state?
2. **Future-operational action meaning:** can two actions be distinguished by some later controlled experiment, even if their immediate outcome laws coincide?
3. **Compositional topology:** do learned action maps close into a stable graph or group-like structure?
4. **Hodological geometry:** do optimal goal costs define a genuine metric, and is that metric Euclidean in low dimension?

Our exact qutrit example answers the first three positively and the fourth negatively. That separation is one of the central results.

2 Controlled quantum experiments from first principles

2.1 States, channels, and instruments

Let $\mathcal{H} \simeq \mathbb{C}^d$. A quantum state is a positive semidefinite operator $\rho \succeq 0$ with $\operatorname{Tr} \rho = 1$. A quantum channel is a completely positive, trace-preserving linear map. A *controlled instrument* for action $a \in \mathcal{A}$ is a finite set

$$\mathfrak{I}^a = \{\mathcal{E}_o^a : o \in \mathcal{O}_a\}$$

of completely positive, trace-nonincreasing maps whose sum $\Phi_a = \sum_o \mathcal{E}_o^a$ is trace preserving [1, 2]. When $\mathcal{E}_o^a(\rho) = K_{a,o}\rho K_{a,o}^\dagger$, $\sum_o K_{a,o}^\dagger K_{a,o} = I$. More general outcomes may have several Kraus operators.

For state ρ, the probability and normalized posterior are

$$p(o \mid \rho, a) = \operatorname{Tr}[\mathcal{E}_o^a(\rho)], \qquad \rho' = \frac{\mathcal{E}_o^a(\rho)}{p(o \mid \rho, a)}.$$

The outcome-forgotten state is $\Phi_a(\rho)$. This distinction between selective and nonselective evolution matters throughout.

2.2 Histories, tests, and the system-dynamics matrix

A history is an observed string

$$h = (a_1, o_1) \cdots (a_t, o_t).$$

Its unnormalized state is

$$\widetilde{\rho}_h = \mathcal{E}_{o_t}^{a_t} \circ \cdots \circ \mathcal{E}_{o_1}^{a_1}(\rho_0), \qquad p(h) = \operatorname{Tr} \widetilde{\rho}_h.$$

A test $\tau = (b_1, y_1) \cdots (b_k, y_k)$ specifies future controls and queried outcomes. Conditional on a possible history,

$$p(\tau \mid h) = \frac{\operatorname{Tr}[\mathcal{E}_{y_k}^{b_k} \circ \cdots \circ \mathcal{E}_{y_1}^{b_1}(\widetilde{\rho}_h)]}{\operatorname{Tr} \widetilde{\rho}_h}.$$

Arrange these probabilities into the generally infinite Hankel or system-dynamics matrix

$$\mathbf{H}_{h,\tau} = p(\tau \mid h).$$

This contains precisely the controlled input–output behavior. A predictive-state representation (PSR) chooses core tests $\tau_1, \dots, \tau_r$ and represents a history by

$$q(h) = \big(p(\tau_1 \mid h), \dots, p(\tau_r \mid h)\big).$$

When the Hankel rank is $r < \infty$, all test predictions are linear functionals of an r-dimensional predictive state [4].

Definition 2.1 (Predictive or causal equivalence)**.** Histories h, h' satisfy $h \sim h'$ when

$$p(\tau \mid h) = p(\tau \mid h') \quad \text{for every admissible future test } \tau.$$

The quotient classes are the agent's operational states.

This definition is stricter than equality under one measurement and weaker than equality of density matrices. Restricted controls may leave distinct density operators operationally indistinguishable. Conversely, a classical memory state attached to the controller may distinguish histories even when the physical density operator does not. Physical Hilbert dimension, predictive dimension, and controller-memory dimension are therefore different resources.

Proposition 2.2 (Finite observability criterion)**.** *For a finite candidate set $S = \{s_1, \dots, s_n\}$, the states are operationally observable if and only if there exists a finite set of tests T such that the row vectors $\big(p(\tau \mid s_i)\big)_{\tau \in T}$ are pairwise distinct.*

Proof. If such a finite set exists, every pair is separated by one of its tests. Conversely, if every pair is predictively inequivalent, choose one separating test for each of the finitely many pairs and take their union. □

2.3 Action equivalence and gauges

An opaque action token has meaning only through how it transforms future predictions. For a predictive state q, define the outcome-labelled update kernel

$$T_a(q; o) = \big(p(o \mid q, a), q_{a,o}\big).$$

Two actions are operationally equivalent if these kernels agree on all reachable predictive states, or equivalently if their shifted Hankel blocks

$$\mathbf{H}^{(a,o)}_{h,\tau} = p(o, \tau \mid h, a)$$

agree for every o, h, τ.

Internal coordinates are not unique. A minimal linear PSR admits similarity transformations $q \mapsto Sq$, accompanied by inverse transformations of test functionals. Gate-set tomography likewise has a similarity gauge that leaves every circuit probability invariant [5, 6]. State labels may be permuted, and a recovered square chart is ambiguous under rotations and reflections. These are not estimation failures. A scientifically meaningful reconstruction must be scored after quotienting by all such operational gauges.

Proposition 2.3 (Finite action-identifiability criterion)**.** *Let S and $\mathcal{A}$ be finite and suppose S is observable. Distinct action meanings are identifiable from controlled statistics if and only if, for every $a \neq b$, there exist $s \in S$, an immediate outcome o, and a finite continuation test τ such that*

$$p(o, \tau \mid s, a) \neq p(o, \tau \mid s, b).$$

Proof. The displayed inequality is a finite experimental witness separating the two shifted Hankel blocks, so it is sufficient. If no witness exists, every admissible experiment has the same law after a and b; no estimator can distinguish them, so it is necessary. □

3 Information, disturbance, and distinguishability

3.1 Several inequivalent notions of informativeness

Let a random variable S index predictive states with prior π_s. For fixed action a, immediate information is

$$I(S;O \mid A=a) = \sum_{s,o} \pi_s p(o \mid s,a) \log_2 \frac{p(o \mid s,a)}{\sum_{s'} \pi_{s'} p(o \mid s',a)}.$$

It vanishes precisely when the current outcome law is state independent. But $I(S;O \mid A=a)=0$ does not imply that the action is meaningless: two actions may prepare different posterior states that a later probe distinguishes.

Local parametric sensitivity can be measured by classical Fisher information,

$$F_a(\theta) = \sum_o p(o \mid \theta, a)[\partial_\theta \log p(o \mid \theta, a)]^2.$$

Global separation of two outcome laws can use total variation, $\mathrm{TV}(p,q) = \frac{1}{2}\sum_o |p_o - q_o|$, or Jensen–Shannon divergence. At the channel level, optimal distinguishability with arbitrary ancillas is governed by the diamond norm $\|\Phi_a - \Phi_b\|_\diamond$ [3]. These quantities answer different questions and should not be collapsed into one score.

Disturbance also requires an explicit reference task. For a pure input, survival fidelity is

$$F_{\mathrm{surv}}(\psi, a) = \langle \psi | \Phi_a(|\psi\rangle\langle\psi|) | \psi \rangle.$$

Trace distance, entanglement fidelity, or loss of future reward may be more appropriate in other experiments. An information–disturbance plot is meaningful only after both axes have been declared.

3.2 Exact no information without universal nondisturbance

Theorem 3.1 (No information without disturbance under a unitary sum)**.** *Let $\{\mathcal{E}_o\}$ be an instrument on a finite-dimensional system. Suppose its nonselective channel equals a fixed unitary channel on the entire operator space:*

$$\sum_o \mathcal{E}_o = \mathcal{U}, \qquad \mathcal{U}(\rho) = U\rho U^\dagger.$$

Then there are probabilities $p_o \geq 0$, $\sum_o p_o = 1$, such that

$$\mathcal{E}_o = p_o \mathcal{U}.$$

Consequently $p(o \mid \rho) = p_o$ is independent of the input state.

Proof. Let $J(\Lambda)$ denote the Choi matrix of a completely positive map. Each $J_o = J(\mathcal{E}_o)$ is positive semidefinite and

$$\sum_o J_o = J(\mathcal{U}) = |U\rangle\!\rangle\langle\!\langle U|,$$

which has rank one. If positive semidefinite matrices sum to a rank-one matrix, every summand has support inside that same one-dimensional support. Hence $J_o = p_o J(\mathcal{U})$ for some $p_o \geq 0$. The Choi isomorphism gives $\mathcal{E}_o = p_o \mathcal{U}$; trace preservation of the sum gives $\sum_o p_o = 1$. Finally $\operatorname{Tr} \mathcal{E}_o(\rho) = p_o \operatorname{Tr} \rho = p_o$. □

Remark 3.2 (Why the assumptions matter). The theorem asserts exact, universal preservation up to one unitary. A quantum nondemolition measurement may reveal which member of an orthogonal or commuting restricted family was supplied. A measurement may preserve an average state without preserving each input. Syndrome extraction may disturb a physical system while preserving an encoded logical subsystem. Approximate preservation permits quantitative information–disturbance tradeoffs. None contradicts Theorem 3.1.

The theorem explains why an integrated spatial action faces tension. To report where the agent is, an instrument must usually disturb some states. To compose as a clean translation, it should preserve the degrees of freedom that carry position except for the intended motion. The design problem is to optimize this tension.

4 From reinforcement learning to goal geometry

4.1 Policies, rewards, and Bellman equations

Reinforcement learning studies sequential decisions. At time t, an agent has an operational state s_t, chooses a_t, observes an outcome and incurs a cost. A policy $\pi(a \mid s)$ specifies action probabilities. For a discounted task,

$$V^{\pi}(s) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t c(s_t, a_t, o_{t+1}) \middle| s_0 = s\right], \qquad 0 \le \gamma < 1.$$

The optimal value satisfies

$$V(s) = \min_a \sum_o p(o \mid s, a)\left[c(s, a, o) + \gamma V(f(s, a, o))\right].$$

For stochastic-shortest-path problems one commonly sets $\gamma = 1$, unit cost per intervention, and terminates upon verified goal success. Properness conditions are then needed to ensure finite values and select the minimal nonnegative Bellman solution [7, 8].

For goal g, let $V_g(s)$ be the optimal expected cost to attain and verify g. A raw directed hodological dissimilarity is

$$C(s, g) = V_g(s) - V_g(g).$$

Subtracting the self-cost is important when verification is noisy and costly. One may symmetrize by

$$D_{sg} = \tfrac{1}{2}[C(s, g) + C(g, s)],$$

but symmetrization alone does not enforce positivity or the triangle inequality.

4.2 Metric and Euclidean tests

A distance matrix D is a metric when it is symmetric, nonnegative, zero only on the diagonal, and obeys $D_{ij} \le D_{ik} + D_{kj}$. To ask whether it is exactly Euclidean, let $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ and form

$$B = -\frac{1}{2} J (D^{\circ 2}) J.$$

Theorem 4.1 (Schoenberg criterion)**.** *A finite symmetric matrix D with zero diagonal is realizable as Euclidean distances among points in $\mathbb{R}^k$ if and only if $B \succeq 0$ and* $\operatorname{rank} B \le k$ *[9].*

Proof. If $D_{ij} = \|x_i - x_j\|_2$ and the coordinates are centered, expanding squared distances gives $B = XX^T \succeq 0$, with rank at most k. Conversely, diagonalize $B = Q\Lambda Q^T$. When $B \succeq 0$, take $X = Q\Lambda^{1/2}$ using its positive eigenvalues. Double centering guarantees that pairwise squared distances between rows of X equal D_{ij}^2. The number of positive eigenvalues is the embedding dimension. □

When exact embeddability fails, classical multidimensional scaling retains the leading positive eigenvalues. Stress summarizes distortion, but a visually pleasing plot is not a proof. In particular, a finite torus has a legitimate two-generator topology while its all-pairs word metric need not embed isometrically in the Euclidean plane.

4.3 Finite necessary-and-sufficient certificate

Criterion 4.2 (Predictive two-dimensional hodology)**.** For finite histories, tests, actions, and goals, a statistics-learnable exact two-dimensional hodological geometry exists if and only if all of the following hold on the operational quotient:

(P) chosen predictive states have distinct finite Hankel rows;
(A) intended action meanings have distinct finite shifted-Hankel blocks;
(B) every goal Bellman equation has a finite minimal nonnegative solution under the declared costs and stopping rule;
(M) the resulting D is a metric;
(E) $B = -\frac{1}{2}JD^{\circ 2}J \succeq 0$ and $\operatorname{rank} B \leq 2$.

If actions are additionally to mean homogeneous local translations, require:

(L) recovered action kernels are local in the embedding and their coordinate displacement laws are state independent up to admitted boundary or gauge symmetries.

Proof. Necessity follows directly: learned states and actions must be statistically separable; values must exist; Euclidean distance is a metric and satisfies Theorem 4.1; homogeneous translations have the stated covariance. For sufficiency, (P) and (A) identify the finite operational model from statistics up to gauge. Solve the Bellman systems by (B), construct D by (M), and use the eigenfactorization in Theorem 4.1 to obtain coordinates in $\mathbb{R}^2$. Condition (L) then assigns learned displacement semantics to actions. Coordinates are unique only up to Euclidean isometry when the distance data are complete and span the plane. □

This criterion is intentionally modular. The qutrit experiment below passes (P), (A), and a topological analogue of (L), but fails (M). That is a sharper diagnosis than reporting a single embedding stress.

5 Qubit case study: an exact predictive action chart

5.1 Instrument and exact probability laws

The first construction asks how much operational semantics a single qubit can carry. Reset the qubit to

$$\rho_0 = |{+x}\rangle\langle{+x}| = \tfrac{1}{2}(I + X).$$

An opaque action b applies a Z-rotation

$$U_{\theta_b} = e^{-i\theta_b Z/2}$$

and then an unsharp binary X-measurement with effects

$$E_o = \tfrac{1}{2}(I + osX), \qquad o \in \{+1, -1\}, \quad 0 < s < 1.$$

A Lüders realization is

$$K_{b,o} = \sqrt{E_o}\, U_{\theta_b}, \qquad \mathcal{E}_o^b(\rho) = K_{b,o}\rho K_{b,o}^\dagger.$$

Completeness follows from $\sum_o E_o = I$. After the rotation, the reset Bloch vector is $(\cos\theta_b, \sin\theta_b, 0)$, so the immediate outcome law is

$$p(+ \mid b) = \mathrm{Tr}(E_+ U_{\theta_b}\rho_0 U_{\theta_b}^\dagger) = \frac{1 + s\cos\theta_b}{2}. \tag{1}$$

Thus the current outcome learns only the cosine of the phase. It cannot distinguish θ from $-\theta$.

The nonselective Lüders measurement dephases perpendicular to the X-axis. Writing

$$\eta = \sqrt{1 - s^2},$$

the outcome-forgotten post-action Bloch vector is

$$r_b = (\cos\theta_b, \eta\sin\theta_b, 0).$$

Two future probes close the chart:

$$p(X+ \mid b) = \frac{1 + \cos\theta_b}{2}, \qquad p(Y+ \mid b) = \frac{1 + \eta\sin\theta_b}{2}. \tag{2}$$

Therefore, for known $s < 1$,

$$\theta_b = \mathrm{atan2}\left(\frac{2p(Y+ \mid b) - 1}{\eta}, 2p(X+ \mid b) - 1\right) \pmod{2\pi}. \tag{3}$$

This is an exact future-operational semantics theorem: two actions with different phases have different finite test rows even if their immediate binary distributions coincide.

Proposition 5.1 (Exact qubit action identifiability)**.** *For $0 < s < 1$, the two probe probabilities in Eq. (2) identify θ_b modulo 2π. If the Y probe is omitted, the action chart is generically only identifiable up to $\theta \leftrightarrow -\theta$.*

Proof. Since $\eta > 0$, the two centered probabilities recover $(\cos\theta_b, \sin\theta_b)$; atan2 gives the phase. With only the X coordinate, cosine is even. □

5.2 Parameter search and finite-sample recovery

A search over 11,799 parameter triples selected

$$\alpha = 0.64, \qquad \beta = 2.02, \qquad s = 0.45, \qquad \eta = 0.8930286.$$

The search balanced several finite-horizon margins: minimum goal separation 0.64 radians, action separation 1.28 radians, nearest alias separation through history length eight 0.10 radians, immediate outcome gap 0.5564, and future-probe gap 0.5333. These are not new fundamental constants; they are experimental design margins chosen to make inference statistically stable while retaining appreciable coherence.

Table 1: Recovery of the qubit action chart from repeated probes. Exact chart recovery is scored modulo the dihedral D_4 gauge of a square. Each row uses 300 independent repetitions of the full reconstruction experiment.

Samples per setting	Exact recovery fraction	Coordinate RMSE
2	0.8733	0.7101
5	0.9867	0.4113
10	1.0000	0.2604
25	1.0000	0.1609
100	1.0000	0.0809
1000	1.0000	0.0261

The approximate $N^{-1/2}$ fall of RMSE is the expected binomial estimation scaling away from singular chart points. In a larger predictive-tomography test, 1,750 histories were probed in X, Y, Z, with 300 samples per probe. The mean reconstructed-state trace distance was 0.0350. This demonstrates finite-horizon observability; it does not claim full device tomography outside the sampled control language.

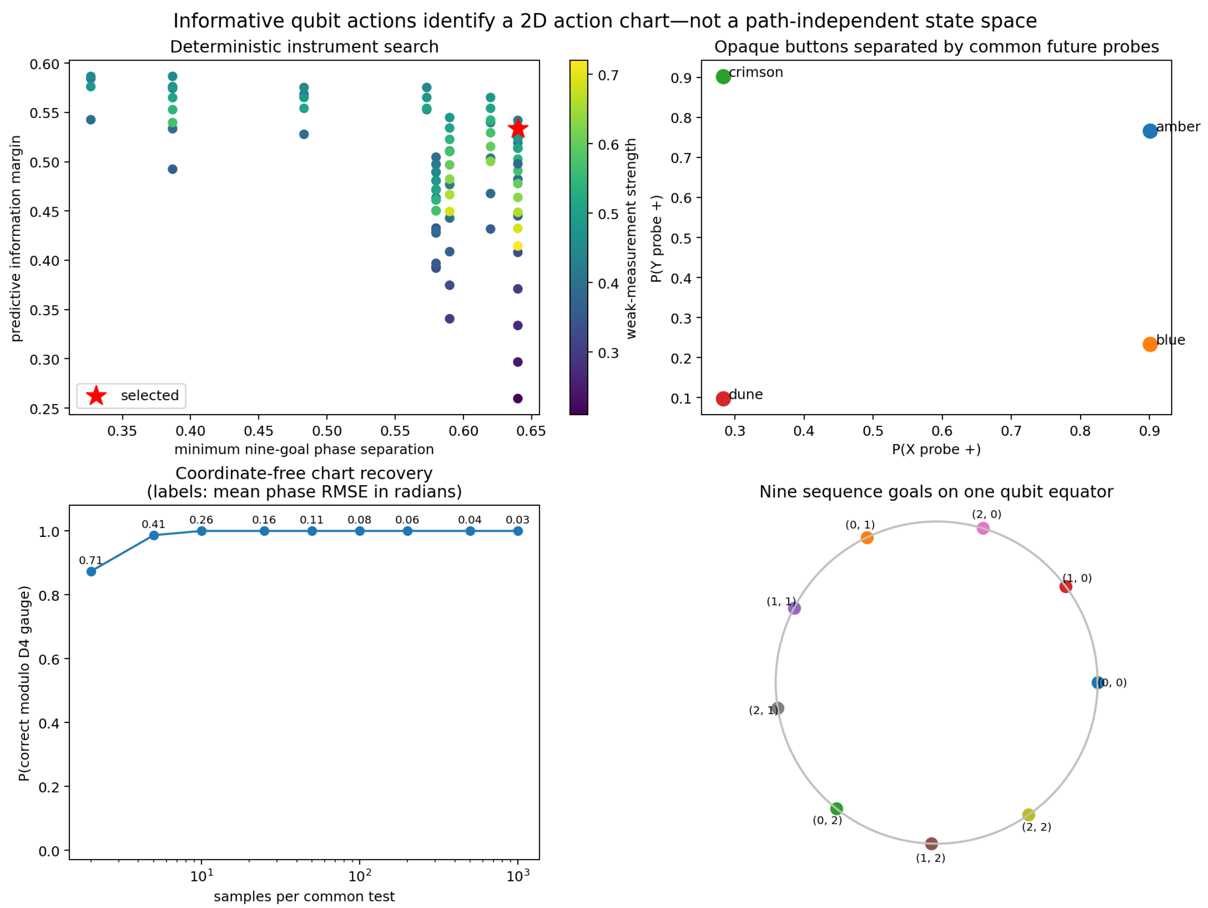

Figure 1: Qubit predictive-semantics experiment. The action's immediate outcome contains cosine information, while future X and Y probes provide an invertible phase chart. Finite-sample reconstruction converges rapidly after quotienting by chart symmetries.

5.3 Decisive controls: meaning need not be immediate

At 200 trials per setting, the informative nonunitary construction recovered the action chart in every replicate. A random-unitary control with a fair, state-independent current coin also achieved 100% action recovery when future probes were allowed: the action changed the future state even though its current outcome carried zero information. A null control achieved 0%. This establishes the logical distinction:

$$I(S; O_t \mid A_t) = 0 \quad \not\Longrightarrow \quad \text{action has no predictive meaning.}$$

The correct test is separation of shifted Hankel blocks, not immediate mutual information alone.

5.4 Why the nonunitary qubit does not furnish an exact plane

An external finite history language can associate words in the two rotations with nine declared displacement classes and thereby reproduce a 3×3 Manhattan chart. That statement is exact for the *language automaton*. It is not exact for the physical nonselective qubit dynamics.

Let D_X be the unsharp-measurement dephasing map on Bloch vectors,

$$D_X(x, y, z) = (x, \eta y, \eta z),$$

and let $R_Z(\theta)$ be the rotation. The actual outcome-forgotten action is

$$\Phi_\theta = D_X \circ R_Z(\theta).$$

For two phases,

$$\Phi_\alpha \Phi_\beta = D_X R_Z(\alpha) D_X R_Z(\beta), \qquad \Phi_\beta \Phi_\alpha = D_X R_Z(\beta) D_X R_Z(\alpha).$$

Unless $\eta = 1$, or angles fall in special degenerate cases, D_X does not commute with R_Z, and these expressions differ. Thus words with the same formal displacement can have different future statistics.

Proposition 5.2 (Compositionality obstruction for the qubit family)**.** *For $0 < s < 1$ and generic nonzero α, β, the nonselective maps Φ_α, Φ_β do not commute. Consequently no quotient that identifies all words solely by the net counts of α and β is an exact predictive-state quotient.*

Proof. In the equatorial plane,

$$D_X R_Z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \eta \sin\theta & \eta\cos\theta \end{pmatrix}.$$

Direct multiplication shows that the commutator has entries proportional to $(1 - \eta)\sin\alpha \sin\beta$ and related generic terms. It is therefore nonzero outside the stated degeneracies. Two word histories related only by exchange of α and β then produce different states, separated by some informationally complete future probe. $\square$

The exhaustive check through word length four makes the size of the failure explicit: among 41 formal displacement classes, 33 were represented by multiple words, none of those 33 was path independent at tolerance 10^{-10}, and the largest within-class predictive probe-vector difference was 0.128490. The remaining eight classes were singletons and hence did not test path independence.

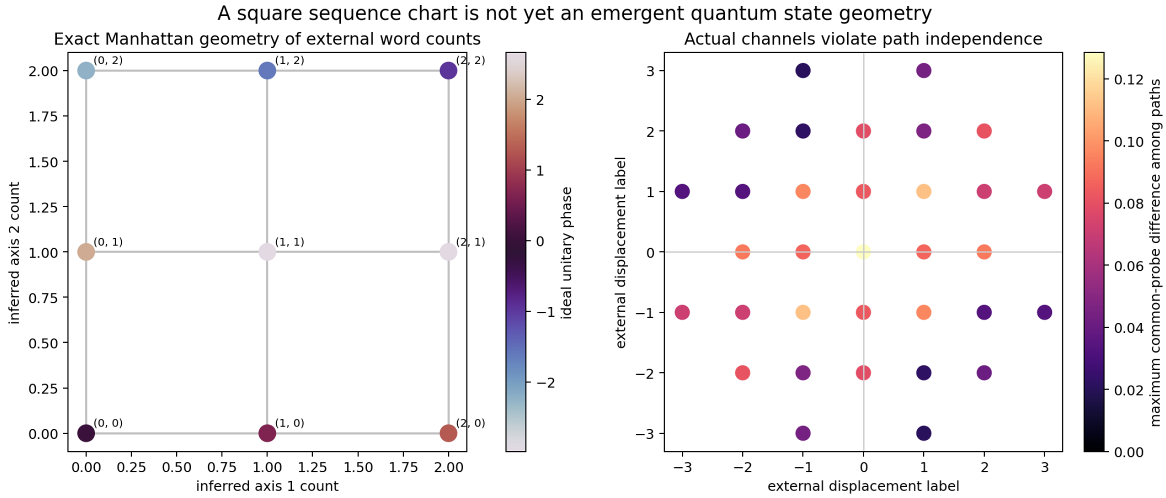

Figure 2: Honest compositionality audit. Words that an external displacement automaton identifies need not be predictively equivalent under the physical nonunitary qubit channels. The learned object is an action chart, not an exact two-dimensional predictive-state space.

The limit $s \to 0$ sends $\eta \to 1$ and restores unitary composition, but also eliminates immediate measurement information. Future probes can still identify actions, yet an integrated instrument that both reports and composes exactly remains unavailable in this family. This is a concrete manifestation of Theorem 3.1.

6 Qutrit case study: a learned Hesse-SIC action topology

6.1 The nine Hesse states

Let $\omega = e^{2\pi i/3}$, and define qutrit Weyl operators

$$X|j\rangle = |j + 1 \bmod 3\rangle, \qquad Z|j\rangle = \omega^j |j\rangle.$$

Starting from

$$|\psi_{00}\rangle = \frac{1}{\sqrt{2}}(0, 1, -1)^T,$$

form the Weyl orbit

$$|\psi_{mn}\rangle = X^m Z^n |\psi_{00}\rangle, \qquad (m, n) \in \mathbb{Z}_3^2,$$

and $\Pi_{mn} = |\psi_{mn}\rangle\langle\psi_{mn}|$. These are the nine Hesse symmetric informationally complete states. They satisfy

$$\frac{1}{3}\sum_{m,n} \Pi_{mn} = I, \qquad \mathrm{Tr}(\Pi_s \Pi_t) = \begin{cases} 1, & s = t, \\ \frac{1}{4}, & s \neq t. \end{cases} \tag{4}$$

The first identity says that $\{\Pi_s/3\}$ is a POVM; the second says all distinct pairs have equal quantum overlap.

The action set is

$$\mathcal{A} = \{I, X, X^\dagger, Z, Z^\dagger\}.$$

Conjugation permutes the SIC orbit. We write

$$U_a \Pi_s U_a^\dagger = \Pi_{T_a(s)}.$$

Although $XZ = \omega^{-1}ZX$ as operators, their conjugation actions commute: the global phase disappears. Therefore the learned transformations can close as translations of $\mathbb{Z}_3^2$.

6.2 One instrument that both moves and reports

Define the integrated instrument

$$K_o^{(a)} = \frac{1}{\sqrt{3}}\Pi_o U_a, \qquad \mathcal{E}_o^a(\rho) = K_o^{(a)}\rho K_o^{(a)\dagger}. \tag{5}$$

Completeness is immediate:

$$\sum_o K_o^{(a)\dagger}K_o^{(a)} = U_a^\dagger\left(\frac{1}{3}\sum_o \Pi_o\right)U_a = I.$$

For an input SIC state $\rho_s = \Pi_s$,

$$\begin{aligned} p_a(o \mid s) &= \mathrm{Tr}[K_o^{(a)}\Pi_s K_o^{(a)\dagger}] && (6)\\ &= \frac{1}{3}\,\mathrm{Tr}[\Pi_o U_a \Pi_s U_a^\dagger] && (7)\\ &= \begin{cases} \frac{1}{3}, & o = T_a(s), \\ \frac{1}{12}, & o \neq T_a(s). \end{cases} && (8) \end{aligned}$$

Moreover, whenever the outcome has nonzero probability,

$$\frac{K_o^{(a)}\Pi_s K_o^{(a)\dagger}}{p_a(o \mid s)} = \Pi_o.$$

Thus the action first translates the likelihood peak, while the emitted token also becomes the exact next predictive state. The last outcome is a sufficient statistic for all future behavior. This measure-and-prepare structure makes the model exactly solvable.

Under a uniform prior on the nine source states, the immediate mutual information is

$$\begin{aligned} I(S;O \mid a) &= \log_2 9 - H\left(\frac{1}{3}, \underbrace{\frac{1}{12}, \ldots, \frac{1}{12}}_{8}\right) && (9)\\ &= 0.251629 \text{ bits.} && (10) \end{aligned}$$

The information is modest but strictly positive.

6.3 Learning topology from opaque tokens

For the learning experiment, both source-state identifiers and action identifiers were independently shuffled. The learner was not supplied the coordinate labels (m, n), the words X, Z, or the direction names. With 2,000 samples for each source–action pair, empirical likelihood tables estimated the peak map

$$\widehat{T}_a(s) = \arg\max_o \widehat{p}_a(o \mid s).$$

All $45 = 9 \times 5$ peak images were recovered correctly. Mean absolute probability error per action lay between 0.00505 and 0.00579.

Purely algebraic diagnostics then found:

- one learned permutation was the identity and had order one;
- the other four had order three;
- all learned permutations commuted;
- they generated a group of order nine;
- the orbit of any token contained all nine tokens.

These are observable signatures of a $\mathbb{Z}_3 \times \mathbb{Z}_3$ regular action, up to permutation and lattice-automorphism gauge. Hence the words horizontal and vertical are interpretations assigned *after* recovery, not labels used to train the estimator.

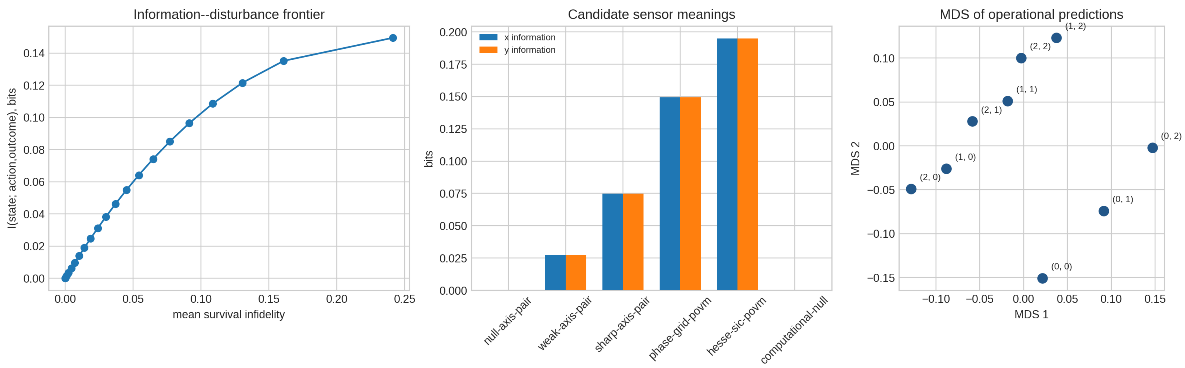

Figure 3: Qutrit candidate and geometry diagnostics. The Hesse-SIC orbit makes every off-diagonal quantum overlap equal, while the action permutations recover a two-generator finite topology. The toroidal word distance has substantial planar MDS distortion and must not be confused with an exact Euclidean metric.

The Cayley graph with generators $X^{\pm 1}, Z^{\pm 1}$ is a 3×3 torus. Its graph distance d_T has shells $0, 1, 2$ with multiplicities $1, 4, 4$. This is unambiguously two-generator and translation covariant. However, two-dimensional metric MDS has stress 0.383. A torus can be topologically two-dimensional without its all-pairs shortest-path metric being a Euclidean planar distance. This is the first warning that recovered topology is weaker than recovered geometry.

7 The exact integrated Bellman obstruction

7.1 Goal protocol and exact solution

Fix a goal token g. Each integrated action costs one. The episode stops when the reported outcome is g; otherwise the next predictive state is the reported SIC state. The undiscounted Bellman equation is

$$V_g(s) = \min_a \left[1 + \sum_{o \neq g} p_a(o \mid s) V_g(o) \right]. \tag{11}$$

Translation and dihedral symmetries imply that the value depends only on the toroidal shell. Write v_0, v_1, v_2 for self, edge, and diagonal, respectively.

At the goal, the identity action centers the likelihood at g. Its success probability is $1/3$; the four edge and four diagonal failures each have probability $1/12$. Hence

$$v_0 = 1 + \frac{1}{3} v_1 + \frac{1}{3} v_2. \tag{12}$$

From an edge state, one translating action maps the premeasurement peak to g, producing exactly the same report kernel as identity at the goal. Because both actions cost one and every failure branch is determined solely by the outcome,

$$v_1 = 1 + \frac{1}{3}v_1 + \frac{1}{3}v_2 = v_0. \tag{13}$$

From a diagonal state, an optimal cardinal move centers the report at a state adjacent to g. Among nonsuccess outcomes, the center and three further edge tokens contribute total coefficient $1/3 + 3/12 = 7/12$, while four diagonal tokens contribute $4/12 = 1/3$. Therefore

$$v_2 = 1 + \frac{7}{12}v_1 + \frac{1}{3}v_2. \tag{14}$$

Solving Eqs. (12) to (14) gives

$$\boxed{v_0 = 4, \qquad v_1 = 4, \qquad v_2 = 5.} \tag{15}$$

Direct enumeration of the five Bellman actions verifies the assumed minimizers and the minimal nonnegative solution.

Proposition 7.1 (Exact self–edge no-go for the report-again Hesse protocol)**.** *The goal-normalized cost $C(s,g) = V_g(s) - V_g(g)$ is zero for every state adjacent to g. It therefore violates identity of indiscernibles and cannot be a metric, regardless of embedding algorithm.*

Proof. Equation Eq. (15) gives $V_g(s) = V_g(g) = 4$ for an edge state. Thus $C(s,g) = 0$ although $s \neq g$. □

Terminal-semantics correction. This proposition concerns the event goal "make the instrument report g again," so another intervention is charged even when the present predictive state is already g. It is not a no-go for state-hitting hodology. For the place-like goal "the current predictive state is g," the Bellman boundary is instead $V_g(g) = 0$. Keeping the same transition kernel, the edge and diagonal equations give $V_{\text{edge}} = 4$ and $V_{\text{diagonal}} = 5$, hence the strict metric shells $(0, 4, 5)$. The covariant-memory successor develops this corrected semantics and adds full-rank retained-memory branches. Thus the calculation above remains exact, but its scope is an event/report cost rather than distance from the present state to a place.

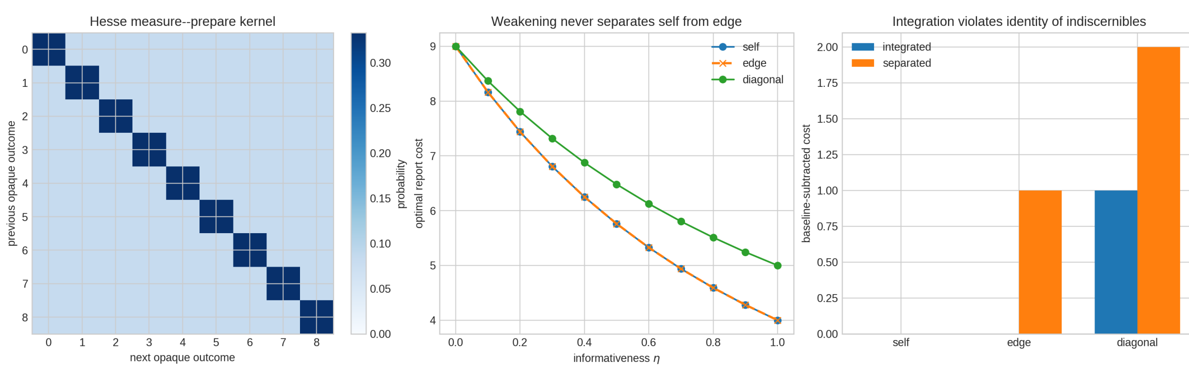

Figure 4: The central negative result. The integrated Hesse-SIC instrument learns a clean opaque action topology, yet exact Bellman analysis yields equal self and edge costs. Weakening the report preserves this equality. The separated move/report benchmark recovers toroidal distance only by giving up integration.

7.2 Why weakening the measurement does not fix it

Mix the exact likelihood with uniform noise:

$$p_a^\eta(o \mid s) = \eta p_a(o \mid s) + \frac{1-\eta}{9}, \qquad 0 \le \eta \le 1. \tag{16}$$

This has a physical measure-and-prepare realization with effects

$$F_o^\eta = \frac{\eta}{3}\Pi_o + \frac{1-\eta}{9} I$$

and posterior preparation Π_o. The same branch-state reset remains.

Table 2: Exact Bellman shell values for the weak integrated family. Numerical rounding is shown to three decimals. Self and edge remain exactly equal throughout the family.

η	self v_0	edge v_1	diagonal v_2
0.0	9.000	9.000	9.000
0.2	7.438	7.438	7.810
0.4	6.250	6.250	6.875
0.6	5.325	5.325	6.124
0.8	4.592	4.592	5.510
1.0	4.000	4.000	5.000

The equality is structural. For an edge state, an aligning move followed by the report has the same outcome kernel as identity followed by the report at the goal. Since (i) both cost one and (ii) every outcome prepares the same Π_o independent of the source, their Bellman action values coincide for every η. Making the measurement weak changes the common value but not the degeneracy. The obstruction is therefore not simply too much measurement; it is the combination of equal move/report cost and complete outcome-only reset.

7.3 Separated motion and report: an exact benchmark, not a solution

As a diagnostic control, allow noiseless unitary translations as actions and a separate Hesse-SIC report action. The agent can first move to g, then report. At g, one report attempt costs one and succeeds with probability 1/3. A failed report prepares one of four edge states or four diagonal states, each with probability 1/12; the mean cost of returning to g is

$$4\left(\frac{1}{12}\right)(1) + 4\left(\frac{1}{12}\right)(2) = 1.$$

If v is the value at the goal,

$$v = 1 + 1 + \frac{2}{3}v, \qquad \text{so} \qquad v = 6.$$

Starting at s, first pay the exact toroidal word distance. Thus

$$\boxed{V_g(s) = 6 + d_T(s, g).} \tag{17}$$

Subtracting the self cost gives exactly d_T. This confirms that the goal protocol and Bellman solver can recover a nondegenerate graph geometry. It does not demonstrate an integrated informative translation: the unitary move emits no spatial information, and measurement is delegated to a distinct action.

8 Computational evidence and controls

The analytic statements above were accompanied by deterministic finite-sample experiments. This section explains what the numerical diagnostics add beyond the proofs.

8.1 Why these metrics were chosen

Kernel mean absolute error tests whether the learned controlled probability law is quantitatively calibrated. Permutation recovery tests the discrete action algebra while respecting token gauge. Held-out negative log likelihood asks whether a learned predictive state generalizes beyond the strings used to identify it. Bellman residual checks that an alleged distance is actually induced by optimal behavior. Identity of indiscernibles is tested before MDS, because no embedding can rescue two distinct predictive states at zero cost. Finally, path closure asks whether the same inferred displacement has the same future law along different routes. These tests address different possible false positives and none can substitute for the others.

For the qubit, 11,799 deterministic parameter triples were searched. With 300 replications, recovery of the opaque two-axis chart was 0.873, 0.987, and 1.000 at two, five, and ten samples per common test. Predictive tomography from 300 samples of each Pauli probe had mean trace-distance error 0.035. Yet zero of 33 repeated-path displacement classes closed predictively, with worst common-probe discrepancy 0.12849.

For the qutrit, 2,000 observations for every source-token/action pair recovered all 45 images of the five hidden permutations. Per-entry controlled-kernel error ranged from 0.00505 to 0.00579. The production bundle also contains 20,000 training and 8,000 held-out five-step sequences and 34,000 navigation episodes. These data verify learnability and prediction; the exact Bellman plateau remains an analytic failure independent of sampling noise.

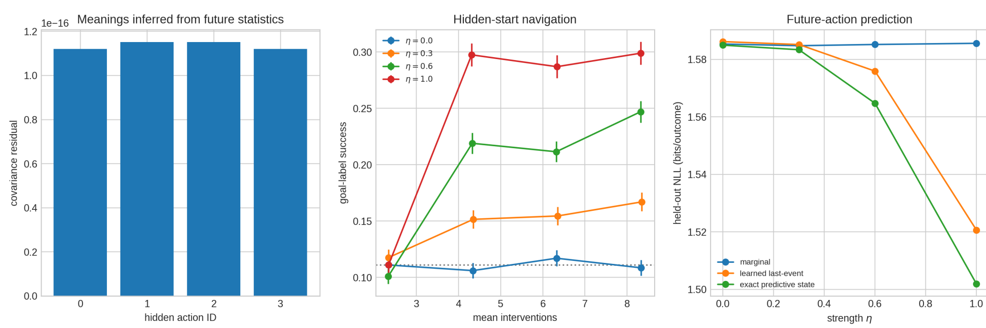

Figure 5: Operational qutrit diagnostics. Opaque actions are inferred from future statistics, informative strength improves hidden-start behavior and held-out prediction, and all evaluation occurs without exposing coordinates to the learner. These positive diagnostics coexist with the exact metric failure in Fig. 4.

8.2 Matched skeptical controls

Four controls locate the source of apparent geometry.

1. A random-unitary catalog has zero immediate mutual information but can still be distinguished by future probes. Thus immediate outcomes are not necessary for operational action semantics.
2. A null quantum channel has rank-zero response variation and supports no quantum action chart.

3. A nine-state external automaton can retain a Manhattan label grid while the quantum state is constant. This shows why word counts are not evidence of predictive quantum space.
4. Independent permutations of actions and outcomes leave all invariant claims unchanged. Coordinate names and compass directions are gauges, not observables.

The controls jointly reject both extremes: neither "the outcome was uninformative, so the action has no meaning" nor "the word graph looked spatial, so space emerged" is valid.

9 What has and has not been established

Table 3: The constructions pass different parts of the operational contract. An entry "future" means that semantics is identifiable by later common tests although the movement outcome itself is null.

Model	Information	Learned structure	Bellman result	Decisive limitation
Pauli qubit square	future	exact four-place action semantics	Euclidean square	movement outcome uninformative
Nonunitary qubit	immediate + future	robust two-axis chart	external Manhattan words	0/33 repeated paths close
Integrated Hesse qutrit	0.251629 bits	exact opaque $\mathbb{Z}_3^2$ topology	self=edge=4, diagonal=5	pseudometric
Separated Hesse qutrit	report only	exact torus translations	$6 + d_T$	movement outcome uninformative
Null quantum DFA	none	external labels only	arbitrary word graph	no quantum predictive geometry

The central positive result is that the meanings of quantum actions can be learned relationally from controlled observations in Hilbert dimension two or three. The Hesse construction further proves that informative integrated actions can reveal an exact two-generator topology. The central negative result is equally precise: in the minimal symmetric measure-and-prepare family, information-producing backaction erases the strict Bellman separation between a goal and its neighbors. Therefore topology, observability, and low-dimensional hodological metric are logically independent requirements.

The no-information theorem is not a universal ban on informative geometric motion. Its assumptions include branchwise pure-state translation and universal nondisturbance on a connected nonorthogonality graph. Mixed-state memory, non-rank-one effects, outcome-conditioned displacement, approximate closure, or an enlarged predictive fiber can evade those assumptions. Such evasions must be charged in the predictive state rather than hidden in a coordinate counter.

10 Most promising next direction

The next experiment should search outcome-conditioned, group-covariant qutrit channels with retained quantum memory. Rather than choosing a geometric target after simulation, optimize one family jointly against the finite certificate:

1. positive immediate mutual information, bounded away from zero;

2. predictive separation and held-out sequence likelihood;
3. identifiable inverse action pairs and low algebra residual;
4. path closure for commuting squares and longer held-out words;
5. a proper Bellman solution with strict $V_{\mathrm{self}} < V_{\mathrm{edge}} < V_{\mathrm{diagonal}}$;
6. a symmetric metric with a positive-semidefinite rank-two Schoenberg Gram matrix, or a controlled approximation whose error is reported;
7. null, cloned-action, random-unitary, episode-rescrambled, and external-DFA controls.

A practical parametrization is to tie Kraus branches covariantly, $\mathcal{E}_o^a = \mathcal{U}_a \circ \mathcal{F}_{T_a^{-1} o}$, while allowing each fiducial map $\mathcal{F}_o$ to have rank larger than one. Rank-one measure-and-prepare maps erase too much memory; higher-rank branches may report partial location while preserving the phase information required for composition. A differentiable inner Bellman solver can evaluate shell margins, while an outer constrained optimizer enforces complete positivity and trace preservation. Candidates must then be relearned from sampled opaque histories by a spectral predictive state model, so success cannot depend on access to their matrices.

This program also offers a disciplined route toward a fiber bundle. The translation quotient would form the base; residual predictive distinctions left after fixing place would form the fiber. Commutator loops of learned action operators could then measure operational holonomy. Curvature and spacetime analogies should be deferred until a flat, informative baseline passes every condition above.

11 Reproduction and data provenance

From the directory containing this paper, run

```
cd qubit
python -m unittest discover -s tests -v
MPLBACKEND=Agg python run_qubit_experiment.py
cd ../qutrit
python -m unittest -v test_informative_qutrit.py
MPLBACKEND=Agg python informative_qutrit.py --episodes 2000 --seed 20260812
```

The qubit and qutrit strands contain seven and nine focused tests, respectively. Saved CSV and JSON files include exact kernels, parameter-search rankings, finite-sample recovery, word-equivalence audits, information– disturbance sweeps, Bellman matrices, held-out prediction, navigation, controls, and complete manifests. Hidden permutations and coordinates are used only for frozen evaluation, never as learner inputs.

12 Conclusion

The user's criticism of dead reckoning changes the standard of evidence in a productive way. A spatial action is not defined by a matrix name, nor by an external counter updated when a button is pressed. Its meaning is the relational transformation it induces on the probabilities of future experience.

Under that standard, a qubit already supports learnable two-direction action semantics, and a qutrit supports an exactly learnable $\mathbb{Z}_3^2$ translation topology with genuinely informative outcomes. But the strongest integrated candidate is not a hodological metric: self and neighbor have equal optimal report cost. No current model in this study simultaneously achieves informative integrated action, predictive path closure, and a nondegenerate exact two-dimensional metric. This honest obstruction supplies the right objective for the next generation of quantum environments.

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